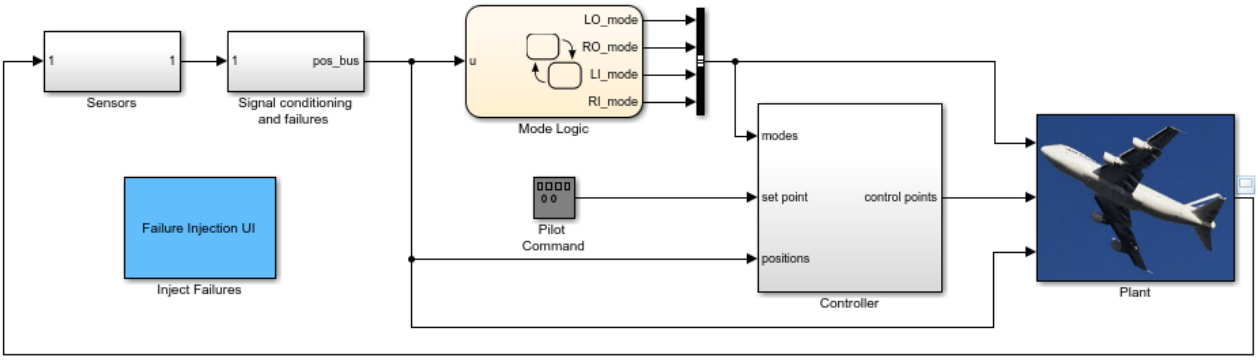
Control of Autonomous Systems – Autumn 2026

Jerome Jouffroy

Exercise Session 1



0. MATLAB/Simulink primer.

0.1. Consider the static system $y = 3\sin(u)$. Compute in MATLAB the value y for different input values u.

0.2. In a script, calculate (using a "for" loop) the values of y for an interval of $u \in [0, 2\pi]$. Plot $y(u)$.

0.3. Given the vector $\mathbf{v} = [1, 2]^\top$, use the expression $\mathbf{w} = \mathbf{R}\mathbf{v}$ (written in MATLAB), where

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}, \tag{1}$$

to compute the *rotated* vector $\mathbf{w}$ (θ is given as $2\pi/3$).

0.4. Just for fun, try the command `why` several times.

0.5. Create a new Simulink file, where you implement the first order differential equation given by $\dot{x}(t) = -3x(t)$. Connect a scope to it and run the simulation for different initial conditions.

1. Simulation of a Mass-Spring-Damper system.

From "Ressources "in Itslearning, download **to your current folder in Matlab** the Simulink file "MSDsystem_temp.slx". After opening Simulink, check that the program works by running the simulation. You should see a simple animation window of an MSD system. Implement in "MSDsystem_temp.slx" the dynamical model of a Mass-Spring-Damper system and connect the output of your block diagram to the

animation and scope. Choose m, k and d such that the position will exhibit damped oscillations when the system is started on non-zero initial conditions.

2. **Two-wheeled robot**. Construct a simple simulation of a two-wheeled robot in the template file "twowheeledrobot_temp.slx".

   2.1. Steer the control inputs with signals you create yourself, such that the robot will make a square trajectory coming back to its original position.

   2.2. Reprogram the same system using this time only one integrator block and the block "MATLAB Function". Requires much less work, right? ;-)

3. **A weird dynamical system (analog computation)**. Consider the following system

$$\dot{\mathbf{x}} = \mathrm{diag}(\mathbf{c})\mathbf{x} - (\mathbf{c}^\top \mathbf{x})\mathbf{x} \tag{2}$$

   where $\dim(\mathbf{c}) = \dim(\mathbf{x}) = n$, $\mathbf{c}$ is a vector/array containing numbers that you choose (these numbers have to be different), and where $\mathbf{x}_0 = \frac{1}{n}[1, ..., 1]^\top$.

   3.1. Choosing $n = 4$, implement (2) in Simulink for a particular vector $\mathbf{c}$.

   3.2. Simulate the behavior of (2) for different vectors $\mathbf{c}$. How does it change the final value of the state $\mathbf{x}(t)$?

   3.3. What does this differential equation compute on $\mathbf{c}$, ie what is the state $\mathbf{x}(t)$ compared to $\mathbf{c}$, after a transient?

   3.4. To dynamic equation (2), add the output equation

$$y = \mathbf{c}^\top \mathbf{x} \tag{3}$$

   What does this compute on $\mathbf{c}$?